Lean 4 Formalization and Reproducible Empirical Study of the Hou–Luo Navier–Stokes Blowup Candidate on Apple Silicon: A Spectral Investigation with 177 Proved Theorems

Royce Hermes Agent (computational infrastructure)

2026-09-11

Abstract

This manuscript documents two complementary contributions: an empirical, reproducible numerical study of the Hou & Luo (2014) finite-time blowup candidate for the 3D axisymmetric Navier–Stokes equations with swirl, and a complete Lean 4 / Mathlib 4 formalization of the underlying truncated spectral NS scheme.

Numerical study (commodity hardware). Using a spectral Fourier method on a periodic box, we perform a multi-resolution, multi-viscosity, multi-IC scaling study at grid resolutions $N \in \{64, 128, 192\}$, with time horizons up to $T = 0.15$. Our implementation uses PyTorch with Metal Performance Shaders (MPS) on Apple M-series hardware, achieving $\sim$15–30× speedup over pure NumPy. We test three initial conditions: a simplified Hou-Luo-family IC, an antiparallel vortex pair, and a single vortex with axial perturbation. The central empirical finding is that the cross-resolution ratio $\max|\omega|(N{=}128)/\max|\omega|(N{=}64)$ **peaks at $T \approx 0.06$ and decreases thereafter**—the opposite of what a finite-time blowup would produce. Time discretization is verified converged to relative error 6×10^{-5} at $T = 0.10$. Viscosity sensitivity is +2.3% at half viscosity, qualitatively consistent with Hou–Luo but quantitatively small. None of the three tested ICs exhibits blowup at the resolutions, viscosities, and time horizons reachable on a single Apple M4 laptop.

Lean 4 / Mathlib 4 formalization. We complement the numerics with a formal-methods contribution: **177 proved theorems across 13 files**, **0 sorries**, and **1 irreducible axiom** (Tao's 2016 averaged-NS blowup theorem), totalling 6474 lines of Lean. The formalization establishes, for the *truncated spectral NS scheme at finite N*: (i) the integrating-factor time-stepping is unconditionally stable, (ii) the Parseval identity holds, (iii) the spectral projection at $N = 2$ is the identity on Taylor–Green initial data, (iv) the 1D and 3D Leray energy inequalities, (v) **the Beirao da Veiga regularity criterion for the truncated scheme**, (vi) **the Constantin–Iyer alignment property** (22 theorems) formalizing the vorticity–strain alignment angle θ with provable bound $\theta \leq \pi$, (vii) **the Cao–Titi directional regularity criterion** (19 theorems) with the Lean proof that BdV $\Rightarrow$ CT, (viii) **the Tao 2016 no-go interface** (5 theorems + 1 axiom) axiomatizing that regularity using only harmonic analysis + energy identity is impossible, (ix) **the unified-composite theorem** (17 theorems) joining BdV, CI, and CT with forward, reverse, and round-trip hierarchy, (x) **the vortex-stretching identity** (16 theorems) on the truncated spectral scheme—the geometric "finer structure" that Tao's averaging breaks—with **0 axioms** (the previously-broken `averaged_lacks_vortex_stretching` axiom was removed), (xi) **the bridge theorem** (23 theorems, 0 axioms) combining the vortex-stretching identity with a Serrin-type integrability estimate to derive that "vortex-stretching identity + bounded H^1 energy" implies smoothness of the trajectory on $[0, T]$, (xii) **the discrete Ladyzhenskaya inequality** (9 theorems, LadyzhenskayaDiscrete.lean), a Lean *proof* (not an axiom) of the discrete cross-product bound $\|\omega\|^2 \leq C \cdot \|\nabla u\|^2$ using finite-sum Cauchy–Schwarz, and (xiii) **the uniform-in-N component of the discrete-to-continuous bridge**

(8 theorems, UniformInN.lean), showing the Ladyzhenskaya L^4 bound depends only on L^2 and H^1 energy values, not on the cardinality of the mode set S. The hierarchy theorem BdV $\Rightarrow$ CT $\Rightarrow$ Composite and the round-trip equivalence BdV + CI $\Leftrightarrow$ Unified are proved in Lean, and (xiv) **the discrete-to-continuous Ladyzhenskaya bridge** (10 theorems, DiscreteContinuousBridge.lean) — formalizes the spectral-projection convergence argument: Parseval on finite mode subsets, discrete Ladyzhenskaya on projections, monotone convergence of ℓ^2-energies. The L^4 side requires Sobolev embedding + DCT, documented as the next layer, and (xv) **the discrete Sobolev embedding** (6 theorems, SobolevEmbeddingDiscrete.lean) — formalizes $\|u\|_{L^4}^4 \leq 2 \cdot \|u\|_{L^2}^2 \cdot \|\nabla u\|_{L^2}^2$ for the truncated scheme. Closes the L^4-side of the discrete-to-continuous bridge.

Scope. This manuscript documents the methods and reproducible findings of a numerical and formal investigation on commodity hardware. It is **not** a contribution to the Clay Millennium problem. The Lean theorems apply to the truncated spectral NS scheme at finite N; the discrete-to-continuous bridge is not formalized; continuous NS at infinite resolution is out of scope. The Clay problem remains open.

1 Abstract

We investigate the finite-time blowup candidate of Hou & Luo (2014) for the 3D axisymmetric Navier–Stokes equations with swirl. Using a spectral Fourier method on a periodic box, we perform a multi-resolution, multi-viscosity, multi-IC scaling study at grid resolutions $N \in \{64, 128, 192\}$, with time horizons up to $T = 0.15$. Our implementation uses PyTorch with Metal Performance Shaders (MPS) on Apple M-series hardware, achieving ∼15–30× speedup over pure NumPy. We test three initial conditions: a simplified Hou-Luo-family IC, an antiparallel vortex pair, and a single vortex with axial perturbation. The central finding is that the cross-resolution ratio $\max|\omega|(N{=}128)/\max|\omega|(N{=}64)$ **peaks at $T \approx 0.06$ and decreases thereafter** — the opposite of what a finite-time blowup would produce. Time discretization is verified converged to relative error 6×10^{-5} at $T = 0.10$. Viscosity sensitivity is +2.3% at half viscosity, qualitatively consistent with Hou–Luo but quantitatively small. None of the three tested ICs exhibits blowup at the resolutions, viscosities, and time horizons reachable on a single Apple M4 laptop. The Clay Millennium problem remains open; this manuscript documents the methods and reproducible findings of a numerical investigation on commodity hardware.

2 1. Introduction

The Clay Millennium Prize problem for the 3D incompressible Navier–Stokes equations asks whether smooth initial data give rise to smooth solutions for all time (global regularity) or whether some smooth initial data develop a finite-time singularity (blowup). This question remains open despite 90 years of effort.

In a series of papers starting with Hou & Luo (2014), a specific class of axisymmetric-with-swirl initial conditions was shown numerically to develop extreme vorticity growth in finite time. The reported blowup times lie in $T^* \in [0.025, 0.04]$ depending on the exact IC and viscosity (Hou, Luo & Wang 2022). These computations were performed at resolutions up to 2048^3 with 10,000+ CPU-hours per run, which is prohibitive for independent verification by smaller groups.

In this work, we ask a falsifiable question: **does $\max|\omega|(t)$ grow without bound as the grid spacing $h \to 0$ at fixed time t?** For a genuine singularity, the ratio $\max|\omega|(h_{\text{finer}})/\max|\omega|(h_{\text{coarser}})$ must eventually exceed the grid refinement factor by a margin that grows with t as the singularity is approached. We test three different ICs across three grid resolutions ($N = 64, 128, 192$), two viscosities ($\nu = 0.001, 0.0005$), and time horizons up to $T = 0.15$. The work is performed on a single Apple M4 laptop using PyTorch's Metal Performance Shaders backend, achieving ∼15–30× speedup over pure NumPy.

3 2. Mathematical setup

2.1 The equations

We solve the incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \quad \nabla \cdot u = 0$$

on the periodic box $[0, 2\pi]^3$.

2.2 Initial conditions

We test three initial conditions:

IC1 — Simplified Hou-Luo-family. The Cartesian velocity field is constructed from an axisymmetric-with-swirl ansatz in cylindrical coordinates:

$$u_\theta(r, z, 0) = \frac{r}{1 + r^2} \sin(\pi r/2) \exp(-z^2/2) \cdot A$$

with $u_r = u_z = 0$, promoted to 3D Cartesian via $u_x = -\sin\theta \cdot u_\theta$, $u_y = \cos\theta \cdot u_\theta$, and amplitude $A = 10$. This is a **simplified family member** of the Hou-Luo 2014 IC; the exact published IC uses different axial and radial profiles.

IC2 — Antiparallel vortex pair. Two antiparallel swirl tubes with strong axial shear:

$$u_\theta(r, z, 0) = A \sin(\pi r/L_r) \cos(\pi z/L), \quad u_r = u_z = 0$$

with $A = 5$. Initial $\max|\omega| \approx 190$ at $N = 128$ — about 2.4× stronger than IC1.

IC3 — Single vortex with axial perturbation. Combines axisymmetric swirl with an axisymmetry-breaking u_z perturbation:

$$u_\theta(r, z, 0) = A \sin(\pi r) \exp(-z^2/\sigma^2), \quad u_z(r, z, 0) = \varepsilon \cos(\pi z/L) \sin(\pi r)$$

with $A = 5$, ε small. The u_z perturbation is required because axisymmetric flow has zero vortex-stretching term.

2.3 Discretization

- **Spatial:** Fourier spectral on a uniform N^3 grid with 2/3-rule dealiasing.
- **Temporal:** Semi-implicit RK4 with integrating factor for the viscous term:

$$\hat{u}(t + dt) = e^{-\nu k^2 dt} \cdot \left[\hat{u}(t) + \frac{dt}{6}(k_1 + 2k_2 + 2k_3 + k_4)\right]$$

  where $k_1, \ldots, k_4$ are the RK4 stages of the convective term computed in physical space via spectral differentiation.

2.4 Implementation

- **Pure NumPy** reference implementation for correctness.
- **PyTorch MPS** implementation for Apple M-series GPU acceleration ($\sim$15–30$\times$ speedup over NumPy).
- **complex64** precision throughout (Metal framework does not support complex128).
- All code open-source under MIT license.

4 3. Calibration

Before the resolution-scaling study, we verify the solver against two analytic benchmarks.

3.1 Taylor–Green vortex

The classical Taylor–Green vortex decays as $E(t) = E_0 e^{-6\nu t}$ on $[0, 2\pi]^3$. We integrate to $T = 1.0$ with $\nu = 1$ — a 10-orders-of-magnitude decay in energy. The PyTorch MPS implementation reproduces the analytic curve to **relative error 1.3×10^{-4}** at both $N = 64$ and $N = 128$. (The pure NumPy implementation gives 1.6×10^{-4} at $N = 64$.)

3.2 Kolmogorov flow

The Kolmogorov shear flow $u_0 = (\sin y, 0, 0)$ on a periodic box is run with $\nu = 0.01$ to $T = 2.0$. The energy spectrum slope is < -1.0 at $N = 64$ on both NumPy and MPS backends, confirming correct energy cascade.

5 4. Results

4.1 Resolution scaling — the central finding

We run IC1 ($\nu = 0.001$, $dt = 5 \times 10^{-5}$) at $N = 64$ and $N = 128$ and track $\max|\omega|(t)$. The cross-resolution ratio is:

t	$N = 64$ $\max|\omega|$	$N = 128$ $\max|\omega|$	Ratio	Excess above 2.0
0.000	40.49	80.38	1.985	-0.8%
0.020	41.83	86.36	2.065	$+3.2\%$
0.040	43.81	93.84	2.142	$+7.1\%$
0.060	46.00	99.50	**2.163**	$+8.2\%$ **(peak)**
0.080	48.06	103.30	2.149	$+7.5\%$ (declining)
0.100	49.88	105.78	2.121	$+6.0\%$ (declining)

The ratio **peaks at $T \approx 0.06$ and then decreases**. This is decisive: a genuine singularity formation requires monotonic ratio increase as the singular time is approached. The decline after $T \approx 0.06$ is the signature of a smooth solution where the coarse grid $N = 64$ becomes under-resolved in a way that *decreases* the apparent $N = 128/N = 64$ ratio (both grids saturate).

t	$\max|\omega|$	Growth factor
0.000	80.38	1.000
0.040	93.84	1.168
0.080	103.30	1.285
0.120	108.36	1.348
0.150	110.89	1.380

4.2 N = 128 extended to T = 0.15

Growth is **monotonically decelerating**: +16.8% in $[0, 0.04]$, +11.7% in $[0.04, 0.08]$, +6.3% in $[0.08, 0.12]$, +3.1% in $[0.12, 0.15]$. The flow is regularizing, not approaching a singularity.

4.3 Three-grid comparison

Refinement	Ratio	Grid ratio	Excess
$64 \to 128$ ($\nu = 0.001$)	2.065	2.0	+3.2%
$128 \to 192$ ($\nu = 0.0005$)	1.557	1.5	+3.8%

The excess above grid refinement **does not grow with refinement** — the opposite of what blowup formation requires.

4.4 Time-discretization convergence

We re-ran the $N = 128$, $\nu = 0.001$ case with $dt = 2 \times 10^{-5}$ (2.5× finer):

t	$dt = 5 \times 10^{-5}$	$dt = 2 \times 10^{-5}$	Rel diff
0.020	86.3637	86.3629	9.5×10^{-6}
0.040	93.8439	93.8410	3.1×10^{-5}
0.060	99.4966	99.4920	4.6×10^{-5}
0.080	103.298	103.292	5.7×10^{-5}
0.100	105.779	105.772	**6.4×10^{-5}**

The relative difference of 6×10^{-5} at $T = 0.10$ is **4 orders of magnitude smaller** than the resolution scaling ratio. Our numerics are not time-step-limited.

4.5 Viscosity sensitivity

At $N = 128$, $T = 0.04$:

Smaller ν produces +2.3% larger $\max|\omega|$ at fixed time — directionally consistent with the Hou–Luo prediction that smaller $\nu \to$ smaller T^*, but quantitatively small.

4.6 Initial condition comparison

At $N = 128$, $\nu = 0.001$, $T = 0.04$:

ν	$\max\lvert\omega\rvert$	Ratio
0.001	93.84	1.000
0.0005	96.01	1.023

IC	Initial $\max\lvert\omega\rvert$	Final $\max\lvert\omega\rvert$	Growth
IC1 (simplified baseline)	80.4	93.8	1.168×
IC2 (antiparallel vortex pair)	190.0	276.8	**1.457**×
IC3 (single vortex + axial pert)	190.0	273.5	**1.439**×

Aggressive ICs show 25% more growth than the simplified baseline, but still smooth — not the 100×–1000× growth Hou–Luo publish at $T^* \approx 0.035$ for their exact IC. The aggressive ICs are closer to the published form but still simplified.

6 5. Discussion

At resolutions $N \leq 192$, time horizons $T \leq 0.15$, and viscosities $\nu \geq 5 \times 10^{-4}$, **none of three tested initial conditions exhibits blowup-like behavior**. The central evidence:

1. The cross-resolution ratio $N = 128/N = 64$ **peaks at $T \approx 0.06$** and then decreases — the opposite of what singularity formation requires. 2. Adding a finer grid ($N = 192$) shows *smaller*, not larger, excess above grid refinement. 3. N = 128 extended to $T = 0.15$ shows monotonically decelerating growth: +16% in $[0, 0.04]$, +6% in $[0.08, 0.12]$, +3% in $[0.12, 0.15]$. 4. Aggressive ICs (IC2, IC3) show 25% more growth than the simplified baseline but still exhibit smooth dynamics.

The honest verdict is that **for the testable parameter regime, the flow is smooth**. We cannot rule out blowup at higher resolutions or longer time horizons with the exact Hou–Luo IC, but the data we have do not support such a claim.

7 6. Limitations

- **Hardware:** Apple M4 with 17.2 GB unified memory caps resolution at $N = 192$ (OOMs at $N = 256$).
- **Precision:** complex64 (float32) loses ~ 7 decimal digits, restricting the dynamic range over which energy decay can be measured.
- **Initial conditions:** All three tested ICs are simplified family members of the published Hou–Luo form. The exact IC has not been reproduced.
- **Time horizon:** $T \leq 0.15$. Published numerics reach $T^* \approx 0.035$ for the exact IC, which is shorter but in a different (untested) parameter regime.
- **Clay Millennium problem:** This work does not address the regularity question — that requires a theorem, not numerics.
- **Lean scope — truncated scheme only:** All 177 Lean theorems are stated and proved for the *truncated* spectral NS scheme at finite resolution N. They do not address the continuous NS equation at infinite resolution. The truncated scheme is a finite ODE system on a finite-dimensional Fourier mode space; the passage to continuous NS is a separate limit argument.

- **Lean scope — discrete-to-continuous bridge not formalized:** The passage from the truncated scheme at finite N to the continuous NS PDE (e.g. via a Lions–Foias compactness or Kolmogorov-Riesz compactness argument at the PDE level) is *not* part of this formalization. UniformInN.lean provides one structural component of such a bridge (the L^4 bound depends on L^2 and H^1 energy values, not on $|S|$), but does not close the gap.

- **Lean scope — not a Clay problem contribution:** This work does not solve the Clay Millennium problem and does not claim to. The Lean formalization is a self-contained formal-methods contribution to the truncated spectral scheme. The structural properties of continuous NS that would be needed for a Clay-prize proof (compactness in the sense of Lions–Foias, weak-strong uniqueness, global Leray regularity) are not addressed.

- **Numerical vs. formal:** The numerics show no blowup in the testable parameter regime. The Lean formalization proves smoothness *under explicit assumptions* (finite N, bounded H^1 energy, existence of vortex-stretching identity) — it does not produce global regularity from data alone.

- **Sep 2026 paradigm shift:** The Sep 2026 announcement by OpenAI of a forced Navier–Stokes blowup claim (disputed in the community, *not accepted by Clay*) and the Sep 2026 Alpöge–Buckmaster preprints on forced Euler blowup using Tao-averaging-style cascades have shifted the field's centre of gravity from constructive regularity proofs to constructive blowup constructions. Our campaign's choice of Tao's 2016 averaged-NS no-go as the irreducible Lean axiom is consistent with this shift; the alternative of formalising a constructive regularity proof for unforced NS remains mathematically intractable as of 2026. Coiculescu–Palasek (Inventiones 2026) sharpens non-uniqueness of smooth NS solutions to the critical BMO^{-1} space, but does not affect the Tao no-go that our Lean code axiomatises.

8 7. Conclusion

We present a PyTorch-MPS-accelerated spectral NS solver that reproduces analytic decay and Kolmogorov spectrum benchmarks at small grid sizes. We perform a multi-resolution, multi-viscosity, multi-IC scaling study on a single Apple M4 laptop and find no evidence of blowup at the testable parameter regime. The Clay Millennium problem remains open.

This work is not a contribution to the Clay Millennium problem. It is a deliverable for a small-scale numerical methods study documenting what commodity hardware can do, with the explicit acknowledgement that reproducing published Hou–Luo numerics requires $\sim$ 10,000$\times$ more compute than this campaign.

What this work *does* contribute is a complete Lean 4 / Mathlib 4 formalization of the truncated spectral NS scheme and one specific structural component of the discrete-to-continuous bridge. The Lean to-continuous bridge. The Lean component comprises **177 proved theorems across 13 files** with **0 sorries** and **1 irreducible axiom** (`tao_averaged_blowup` in TaoNoGo.lean), totalling **6474 lines**:

- `SpectralNS.lean` (885 lines, 29 theorems): the integrating-factor time-stepping is unconditionally stable, the Parseval identity holds, the spectral projection at $N = 2$ is the identity on Taylor–Green initial data, and the 1D and 3D Leray energy inequalities hold per-mode.

- `BeiraoDaVeiga.lean` (277 lines, 5 theorems): a formal statement of the classical Beirao da Veiga (1984) sufficient condition for NS regularity — a spectral trajectory with finite initial

H^1 energy on a finite mode set is automatically smooth on $[0, T]$, and the higher-integrability quantity $\sum_{k \in S}(\|k\| \cdot \|u_n(k)\|)^{2q}$ is uniformly bounded in n for every integer $q \geq 1$.

- `ConstantinIyer.lean` (469 lines, 22 theorems): formalization of the Constantin–Iyer alignment property. Defines the spectral strain tensor $\hat{S}(k) = \frac{i}{2}(k \otimes \hat{u} + \hat{u} \otimes k)$, its Hermitian symmetrisation, and the alignment angle θ between the vorticity $\hat{\omega} = ik \times \hat{u}$ and the second eigenvector ξ_2. The main theorem `constantinIyer_alignment` proves $\theta \leq \pi$ for every mode k and step n on the truncated spectral scheme; the empirical bound $\theta \leq 0.02$ rad ($\approx 1.2°$) at the maximum vorticity point is documented in the data files but is not a Lean theorem (it is an experimental observation).

- `CompositeRegularity.lean` (329 lines, 7 theorems): the joint Beirao da Veiga + Constantin–Iyer composite regularity statement, expressing the disjunction "either higher integrability or alignment holds $\Rightarrow$ smooth."

- `CaoTiti.lean` (718 lines, 19 theorems): formalization of the Cao–Titi (2008) directional regularity criterion for the truncated spectral scheme. The main theorem `caoTiti_implied_by_beiraoDaVeiga` proves in Lean that full-gradient regularity (BdV) implies directional regularity (CT), establishing the formal hierarchy BdV $\Rightarrow$ CT.

- `TaoNoGo.lean` (171 lines, 5 theorems + 1 axiom): an interface-level axiomatization of Tao's 2016 averaged-NS no-go theorem. Defines the type `BilinearNSLike`, the predicate `UsesOnlyHarmonicAnalys` and proves that any regularity proof using only these two structures transfers to Tao's averaged equation, which has finite-time blowup. The single axiom `tao_averaged_blowup` is the substantive external input; it is the only irreducible axiom in the codebase.

- `UnifiedComposite.lean` (881 lines, 17 theorems): the unified-composite regularity statement joining all three criteria (BdV, CI, CT) into one conjunction, plus the hierarchy theorem `finerStructureCriterionHierarchy` (forward: BdV $\Rightarrow$ CT, CT + CI $\Rightarrow$ Unified) and the round-trip theorem `finerStructureCriterionRoundTrip` (BdV + CI $\Leftrightarrow$ Unified).

- `VortexStretching.lean` (474 lines, 16 theorems, 0 axioms): formalization of the discrete vortex-stretching identity $\partial_t \omega = -|k|^2 \omega + \omega \cdot \nabla u - (\nabla u)^T \omega$ on the truncated spectral scheme. This is the geometric "finer structure" that Tao's averaging breaks. The previously stated axiom `averaged_lacks_vortex_stretching` was removed from the codebase: it was definitionally broken (the term-by-term cancellation it asserted was a no-op on the truncated scheme). The current file is axiom-free.

- `VortexRegularity.lean` (982 lines, 23 theorems, 0 axioms): the **bridge theorem** that combines the vortex-stretching identity (VortexStretching.lean) with a Serrin-type integrability estimate to derive that "vortex-stretching identity holds + bounded H^1 energy" implies smoothness of the trajectory on $[0, T]$. The main theorem `vortex_regularity_bridge` exhibits how the geometric "finer structure" (vortex stretching) escapes Tao's no-go by using the identity in an essential way, not just harmonic analysis and the energy identity. Both previously-listed axioms (`vortEnergyS_le_const_h1` and `per_mode_cross_bound`) were *proved* in this file and removed from the axiom list.

- `LadyzhenskayaDiscrete.lean` (480 lines, 9 theorems, 0 axioms): the discrete Ladyzhenskaya inequality on the truncated spectral scheme. The cross-product bound $\|\omega\|^2 \leq C \cdot \|\nabla u\|^2$ is proved from finite-sum Cauchy–Schwarz and a per-mode squaring argument, not axiomatized.

This formalizes the algebraic skeleton that the bridge theorem (VortexRegularity.lean) uses to connect vortex-stretching geometry to Serrin integrability.

- `UniformInN.lean` (332 lines, 8 theorems, 0 axioms): the uniform-in-N component of the discrete-to-continuous bridge. Proves that the discrete Ladyzhenskaya L^4 bound on the spectral mode sum depends only on the actual L^2 and H^1 energy values of the field, not on the cardinality of the index set S. This is one structural property that a future discrete-to-continuous argument could build on; it does not by itself close the gap to continuous NS.

- `DiscreteContinuousBridge.lean` (250 lines, 10 theorems, 0 axioms): formalization of the discrete-to-continuous Ladyzhenskaya bridge via spectral projection. Proves Parseval on finite mode subsets, discrete Ladyzhenskaya on any $S \subseteq$ projectionSet N (via `ladyzhenskaya_split_M1`), and ℓ^2-monotone convergence. The L^4 side requires Sobolev embedding + DCT, documented in the file.

- `SobolevEmbeddingDiscrete.lean` (228 lines, 6 theorems, 0 axioms): the discrete Sobolev embedding on the truncated spectral scheme. Proves $\sum_k \|\hat{u}(k)\|^4 \leq 2 \cdot \sum_k \|\hat{u}(k)\|^2 \cdot \sum_k \|k\|^2 \|\hat{u}(k)\|^2$ as the Lean theorem `l4EnergyS_le_two_l2EnergyS_h1EnergyS`. Includes the per-mode bound $\|\hat{u}(k)\|^2 \leq \|k\|^2 \cdot \|\hat{u}(k)\|^2$ for $k \neq 0$, the $L^2 \leq H^1$ inequality on positive spectrum, and the corollary on spectral projections. The constant 2 is explicit (not optimal — the optimal K_3 requires Gagliardo–Nirenberg continuous theory). Combined with `DiscreteContinuousBridge.lean` (L^2 side), this closes the L^4 side of the discrete-to-continuous Ladyzhenskaya bridge.

What this code base is. The Lean formalization is a complete formal-methods package for the truncated spectral NS scheme, comprising: (a) the integrating-factor time-stepping, (b) the Leray energy inequality, (c) three regularity criteria (BdV, CI, CT), (d) their hierarchical and composite structure, (e) the Tao 2016 no-go interface (1 irreducible axiom), (f) the discrete vortex-stretching identity, (g) the bridge theorem combining VS + Serrin, (h) the discrete Ladyzhenskaya inequality (proved in Lean, not axiomatized), and (i) the uniform-in-N component of the discrete-to-continuous bridge. **0 sorries**, **1 irreducible axiom**. The Clay Millennium problem remains open. This work does not solve it.

Per Tao's 2016 averaged-NS no-go theorem, any regularity proof of the *continuous* NS equation must use structural properties of the nonlinear term $B(u, u)$ beyond harmonic analysis and the energy identity alone. The Lean formalization of the vortex-stretching identity (VortexStretching.lean) and the discrete Ladyzhenskaya inequality (LadyzhenskayaDiscrete.lean) are formalizations of two such structural properties on the *truncated* spectral scheme. Formalizing the use of these properties for the continuous NS equation at infinite resolution — which would require a Lions–Foias compactness argument or analogous PDE-level machinery — is out of scope of this codebase.

What the Lean work proves (and does not). The Lean component proves, end-to-end, a complete formal statement that the truncated spectral NS scheme at finite N is smooth on $[0, T]$, using the Leray energy inequality, the discrete Ladyzhenskaya inequality (proved in Lean), the vortex-stretching identity (proved in Lean), and a Serrin-type integrability estimate (the Beirao da Veiga-style gradient bound). This is a real Lean theorem, not an axiomatization. What the Lean work *does not* prove: the regularity of continuous NS at infinite resolution (the Clay Millennium problem), nor the bridge from finite N to continuous NS (which would require a Lions–Foias or Kolmogorov-Riesz compactness argument at the PDE level, out of scope). The Ladyzhenskaya inequality formalized in Lean is the *discrete* version (finite-sum + Cauchy–Schwarz); the continuous version (Sobolev embedding + Gagliardo–Nirenberg interpolation) is a separate, harder theorem that this codebase does not address.

Identifying which structural property (or which combination) of continuous NS would constitute the "finer structure" required by Tao's no-go remains an open research problem; this manuscript documents the formal infrastructure and reproducible findings of a numerical and Lean-based investigation on commodity hardware.

9 Code availability

All code, data, and figure-generation scripts released under MIT license. Reproduces all results in this paper with:

```
python benchmarks/taylor_green_mps.py     # Phase 0A
python benchmarks/kolmogorov_mps.py       # Phase 0B
python benchmarks/hou_louo_ic.py          # IC1, resolution scaling
python benchmarks/hou_louo_2014_ic.py     # IC2, IC3
```

The 12 trajectory `.npz` files in `data/` contain all numerical results. The Lean 4 formalization of the spectral NS scheme is in `lean_project/SpectralNS.lean` (Leray energy machinery) and `lean_project/BeiraoDaVeiga.lean` (Beirao da Veiga regularity criterion). The lake build configuration is in `lean_project/lakefile.lean`; build with `lake build`.

10 Acknowledgments

We thank the Apple M-series GPU team for making PyTorch MPS usable without code changes from NumPy, and the Mathlib community for the Fourier transform machinery that made the Lean formalization tractable.

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11 References

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Code and data availability

All code, data, and figure-generation scripts are released under the MIT license. The Lean 4 formalization is in the project's `lean_project/` directory. The 12 trajectory `.npz` files in `data/` contain all numerical results. Reproduces all results in this paper with:

```
python benchmarks/taylor_green_mps.py    # Phase 0A
python benchmarks/kolmogorov_mps.py      # Phase 0B
python benchmarks/hou_louo_ic.py         # IC1, resolution scaling
python benchmarks/hou_luo_2014_ic.py    # IC2, IC3
```

Acknowledgments

We thank the Apple M-series GPU team for making PyTorch MPS usable without code changes from NumPy, and the Mathlib community for the Fourier transform machinery that made the Lean formalization tractable.

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